



HOUSE OF REPRESENTATIVES
H.B. No. 3382

INTRODUCED BY HONORABLE MARK A. VILLAR

EXPLANATORY NOTE

In this age of globalization, science and technology (S&T) have undeniably become the most significant tool in economic growth and source of competitive advantage.

The past years, the Philippines has been silently reaping laurels in science and technology. In 2006, in the World Robotics Olympiad in Nanning, China, the country's representatives Kim Marco Perez, Dann Joseph Garcia, and Francisco Gabriel Nunez of the First Asia Institute of Technology and Humanities bagged the Gold Medal in the Open Category with their theme "Humanoids". Equally outstanding are the Winning feats of Filipino students in the Intel International Science and Engineering Fair (ISEF) 2006.

Last year, the Philippine contingent bagged two of the highest awards during the 2012 World Robotics Olympiad held in Kuala Lumpur, Malaysia in November. Dr. Yanga Colleges Inc. of Bocaue, Bulacan won the gold medal for the Junior High School Open Category with their robot creation "HERO." "HERO" or Humanoid for Educational Reinforcement Operation is a teaching assistant robot inspired by the 2009 CNN Hero of the Year, the Philippines' very own Efren Peñaflorida.

It would appear that more and more Filipino students, both in the public and private schools, are gaining interest with robotics, and the Philippine Robotics Olympiad is an annual competition which aims to offer an opportunity for students to explore robots and robotic systems in schools. Every year, grand winners of the PRO represent the country in the World Robot Olympiad.¹

According to the Department of Science and Technology, robotics is one of the platforms the country can depend on. Last 2012, the DOST launched the Smarter Philippines campaign, a program that will use Information and

¹ <http://www.rappler.com/science-nature/32128-12th-philippine-robotics-olympiad>

Communication Technology (ICT) as “an enabling tool to provide more efficient and reliable services in some priority industries in the country”. Robotics can be used in areas like manufacturing, health, agriculture, and so on.²

Thus, this bill is proposed with the foremost consideration of supporting modern education through the integration of robotics application in secondary science curriculum.

Under the bill, robots, as may be applicable, shall be utilized as technology tools that will allow students to perform experimental applications in designing, building and testing gadgets for instructional purposes. They shall be utilized by teachers as complementing their efforts.

Initially, all campuses of the Philippine Science High School and selected science high schools under the Department of Education shall be introduced the knowledge of robotics application. Teacher training that includes but not limited to software programming; robot assembly; integration of robotic concepts on energy, light and optics, electronics and magnetism, force, power, work and motion, shall be robustly conducted.

Science and Technology are what the country needs to survive the modern world. And hence, early enactment of this legislation is earnestly sought.



MARK A. VILLAR

² Ibid.

SIXTEENTH CONGRESS OF THE)
REPUBLIC OF THE PHILIPPINES)
First Regular Session)

HOUSE OF REPRESENTATIVES

HOUSE BILL NO. **3382**

Introduced by HONORABLE MARK A. VILLAR

AN ACT

**INTEGRATING ROBOTIC APPLICATION IN THE TEACHING OF SECONDARY
SCIENCE EDUCATION IN THE PHILIPPINES, APPROPRIATING FUNDS
THEREFOR AND FOR OTHER PURPOSES**

*Be it enacted by the Senate and the House of Representatives of the Philippines in
Congress assembled:*

SECTION 1. The use of modern technology through robotics application is hereby promoted as part of the Philippine science educational curriculum. Robotics application shall be integrated in the teaching of physics and other science subjects, concepts, theories and principles. Robots, as may be applicable, shall be utilized as technology tools that will allow students to perform experimental applications in designing, building and testing gadgets for instructional purposes.

Initially, all campuses of the Philippine Science High School and selected science high schools under the Department of Education shall be introduced the knowledge of robotics application. Teacher training that includes but not limited to software programming; robot assembly; integration of robotic concepts on energy, light and optics, electronics and magnetism, force, power, work and motion, shall be conducted.

SEC. 2. The sum of One Hundred Million Pesos (Php100,000,000.00) is hereby appropriated to support the initial implementation of this Act which shall

include the designing of the course on robotics application, teacher training, development of tools and instructional materials and other concepts on robotics.

SEC. 3. The Department of Science and Technology (DOST) in coordination with the Department of Education (DepEd) and in consultation with all relevant sectors engaged in education shall issue the necessary implementing rules and regulations of this Act.

SEC. 4. This Act shall take effect fifteen (15) days after its complete publication in the Official Gazette or in at least two (2) newspapers of general circulation.

Approved,